

Read the text and answer the exam questions in your book:

If the immune system has encountered a particular pathogen before, it can respond very quickly to a threat. The immune system recognises pathogens that it has encountered before because the antigens in the pathogen's cell membrane stay the same.

If an antigen is recognised, cells of the immune system called memory cells are able to rapidly produce large numbers of antibodies to the pathogen and disable or destroy it before it is able to take hold in the body. When this occurs, the person is said to be immune to that pathogen.

- 1) What advantage does having memory cells in the immune system give to the body?**
- 2) Explain what is meant when someone is said to be immune to a disease. Why do you think it is possible to have immunity against some illnesses and not others?**

Answers

- 1) Your immune system has memory cells to respond rapidly to the invasion of that disease.
- 2) Immune system able to rapidly produce memory cells; destroys pathogen before it causes harm

Some diseases or illnesses you cannot become immune to because they are able to mutate and have different antigens that are not recognised by the memory cells.

Building immunity

Learning Objective:

1. Recall how vaccinations prevent infection.
2. Explain how mass vaccination programmes reduce the spread of a disease.
3. Evaluate the global use of vaccination.

Keywords

immunity, vaccination, vaccine



Queen
Elizabeth
School

What do you think the difference between painkillers and antibiotics is?

Can you list the uses of painkillers and antibiotics?

How do Painkillers work?

They stop the sensory nerve endings from sending pain messages to your brain.



Human Defence Systems

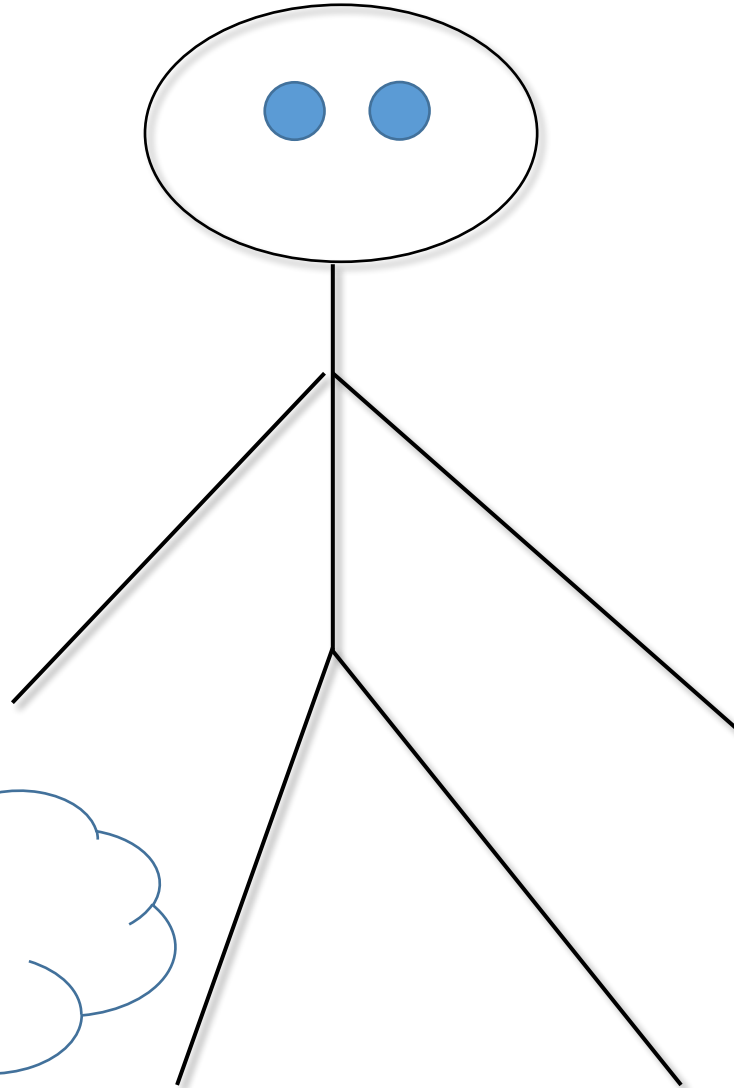
Monday, 08
September 2025

Sometimes we get ill.

Most of the time though, we aren't.

You are constantly covered in germs, bacteria etc

So how does our body stop us from being infected?



For the brave!
How do these areas
stop infection?

Eyes:

- Lashes to trap pathogens
- Tears contains lysozymes; enzymes which destroy bacterial cell walls

Stomach:

- Contains hydrochloric acid (pH 2) which denatures the enzymes
- We can vomit to get rid of pathogens

Respiratory system:

- Cilia to sweep pathogens out into the stomach

Skin:

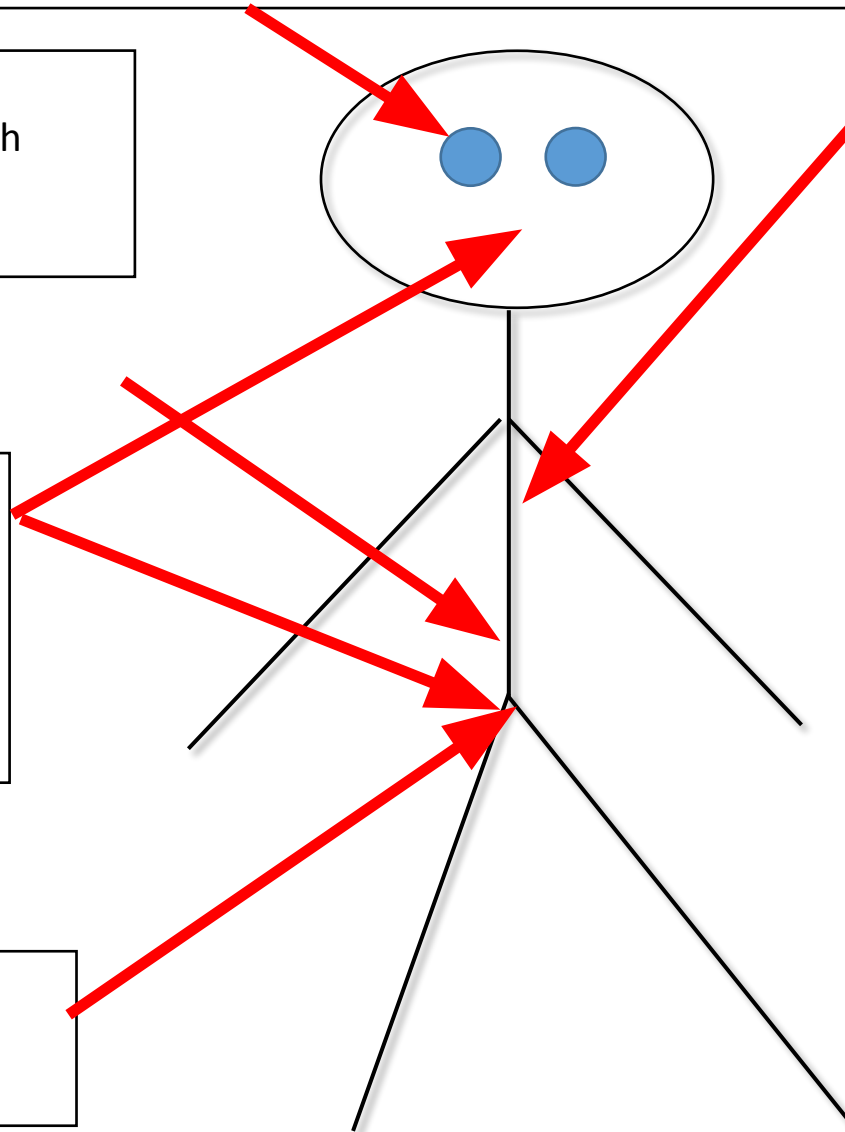
- Physical barrier against infection
- Platelets cause blood clotting if barrier is broken.
- Glands make the skin too acidic for some pathogens
- Skin flora – human skin has a community of harmless microbes on skin which cause competition

Mucous membranes:

- Lines openings (*trachea, nose, vagina, urethra*)
- Mucus contains lysozymes; enzymes which destroy bacterial cell walls
- Mucus is sticky and can trap pathogens

Urinary system:

- Urinary tracts are flushed with urine which has antibacterial properties



Vaccination

Monday, 08
September 2025

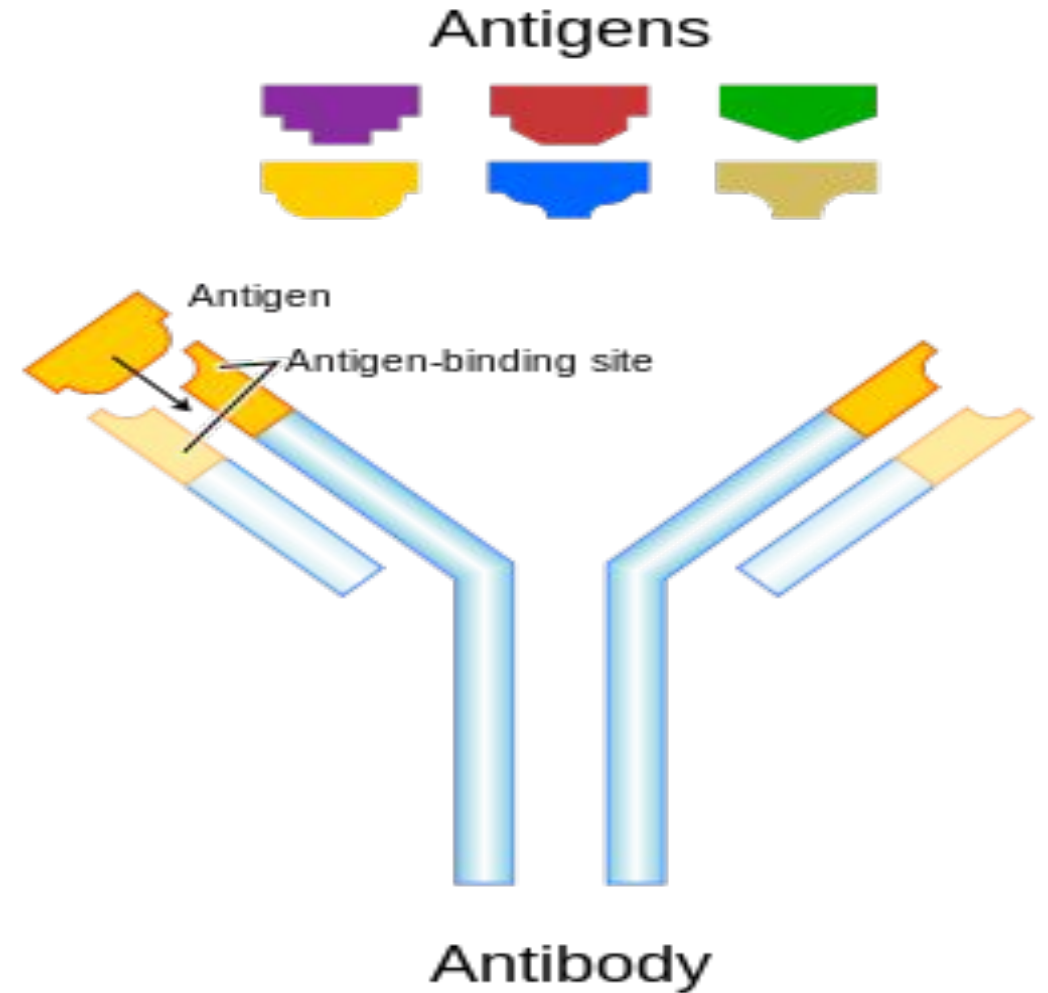
Vaccines cause your body to create antibodies to fight off a disease.

This means that if you were to ever to contract the disease, you would already be immune to it. Your body would possess the antibodies necessary.

You can either get active or passive immunity

Active means YOUR body is making the antibodies

Passive means you are given them. Either in an injection or by e.g. your mother during birth



Task 1: Which of the following diseases can we not be vaccinated against?

- Measles
- Chicken pox
- Rubella
- Tetanus
- Typhoid fever
- Mumps
- Influenza
- Human papillomavirus infection
- Ebola
- Whooping cough (Pertussis)

Small pox Vaccination video

<https://www.youtube.com/watch?v=yqUFy-t4MIQ>

Answer the following question with information from the video,
What is the difference between a vaccination and a variolation?

Variolation - First method used to immunize an individual against smallpox with material taken from a patient.

Vaccination - A dead or weakened form of the disease.

Vaccination

Monday, 08
September 2025

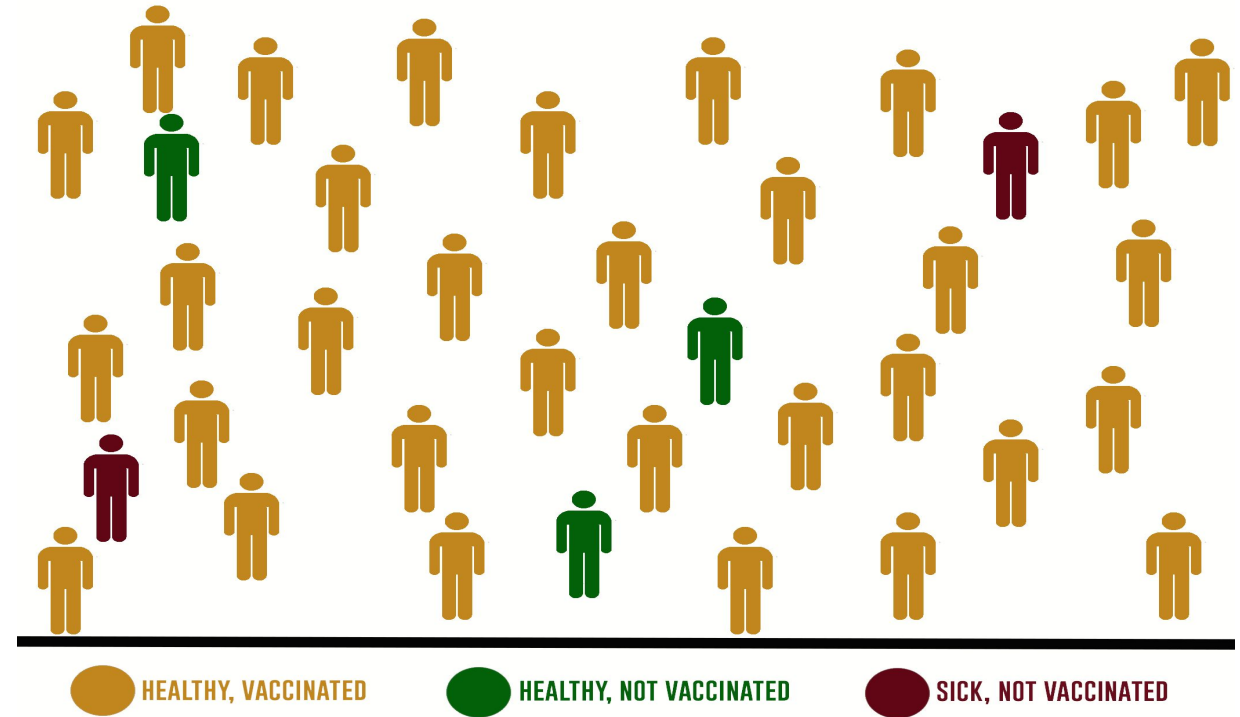
What links this picture to vaccination?

Herd immunity is a concept whereby not everyone needs to be vaccinated to get rid of the disease.

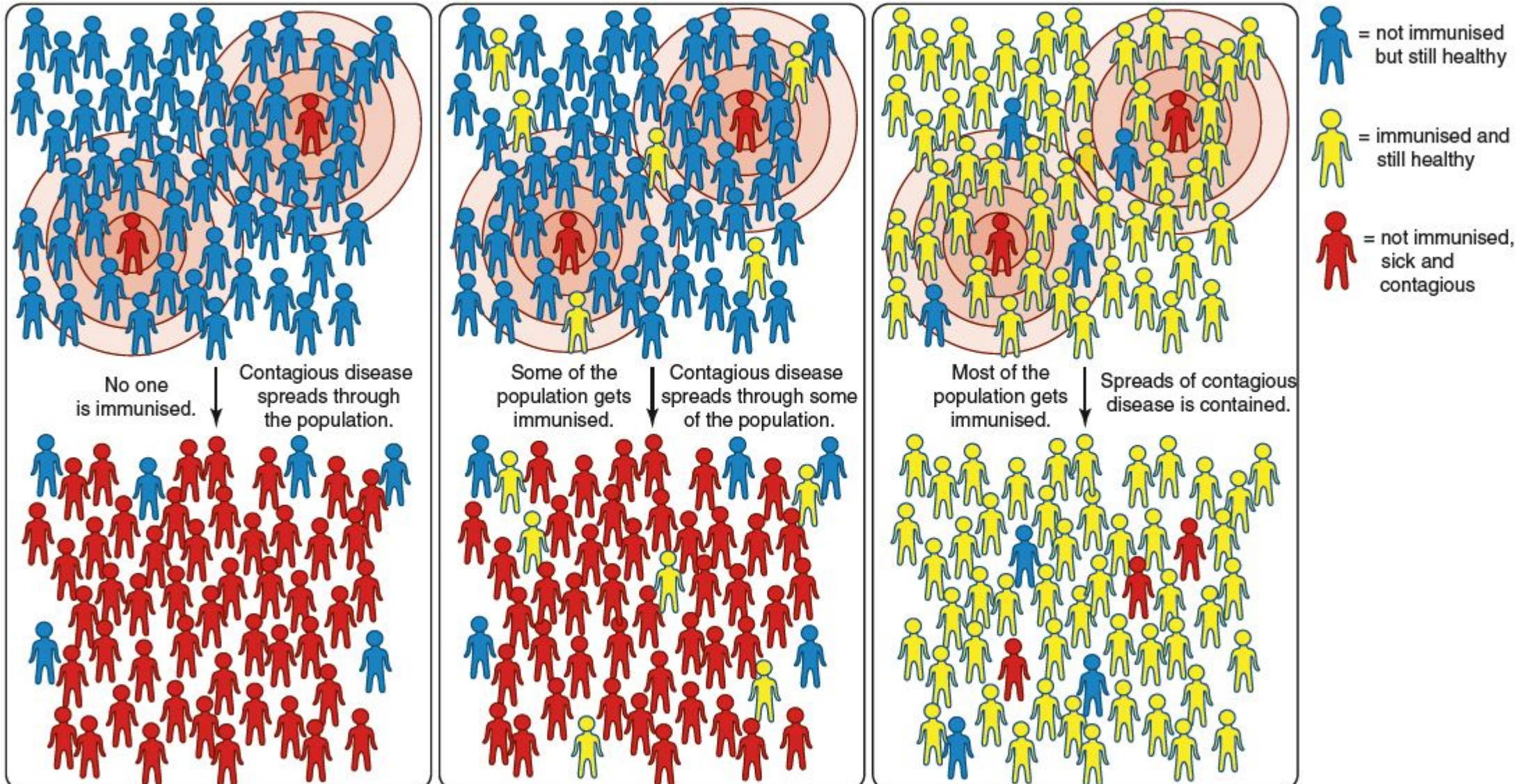
Just enough that it can't get transferred easily enough to survive

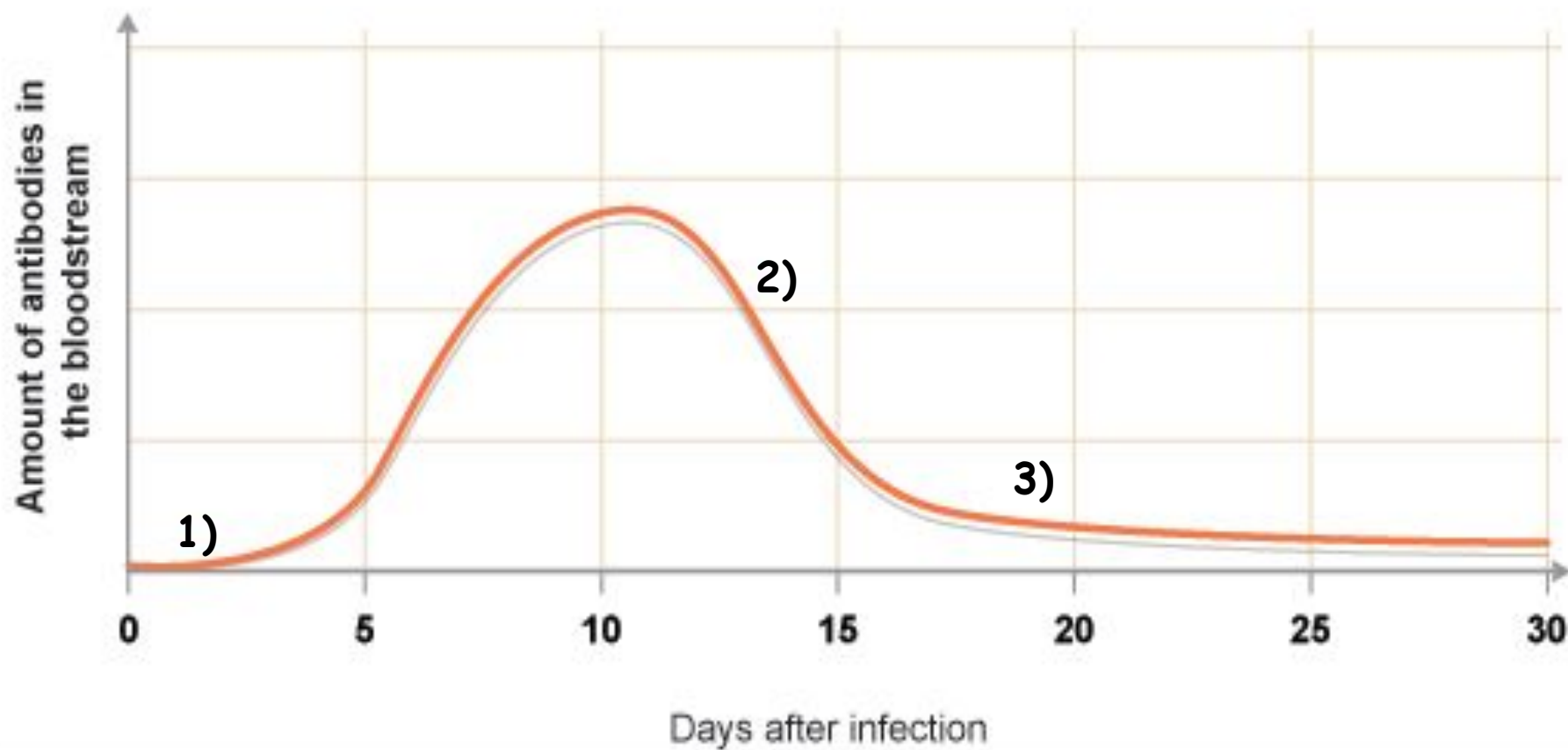
It's like getting rid of most vehicles. We'd struggle to get where we need to go

WHEN ENOUGH PEOPLE ARE VACCINATED



2. How do mass vaccination programmes reduce the spread of disease?

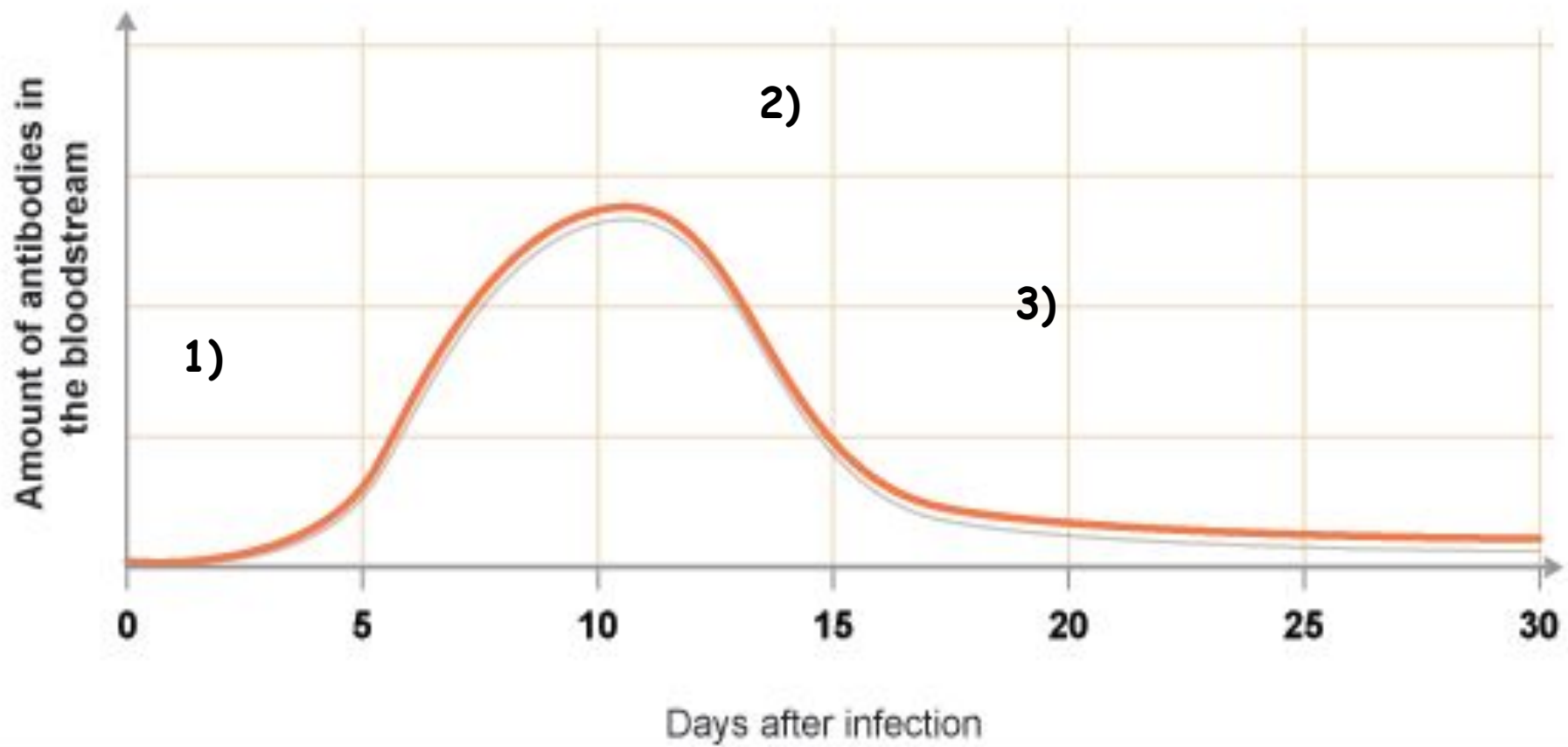


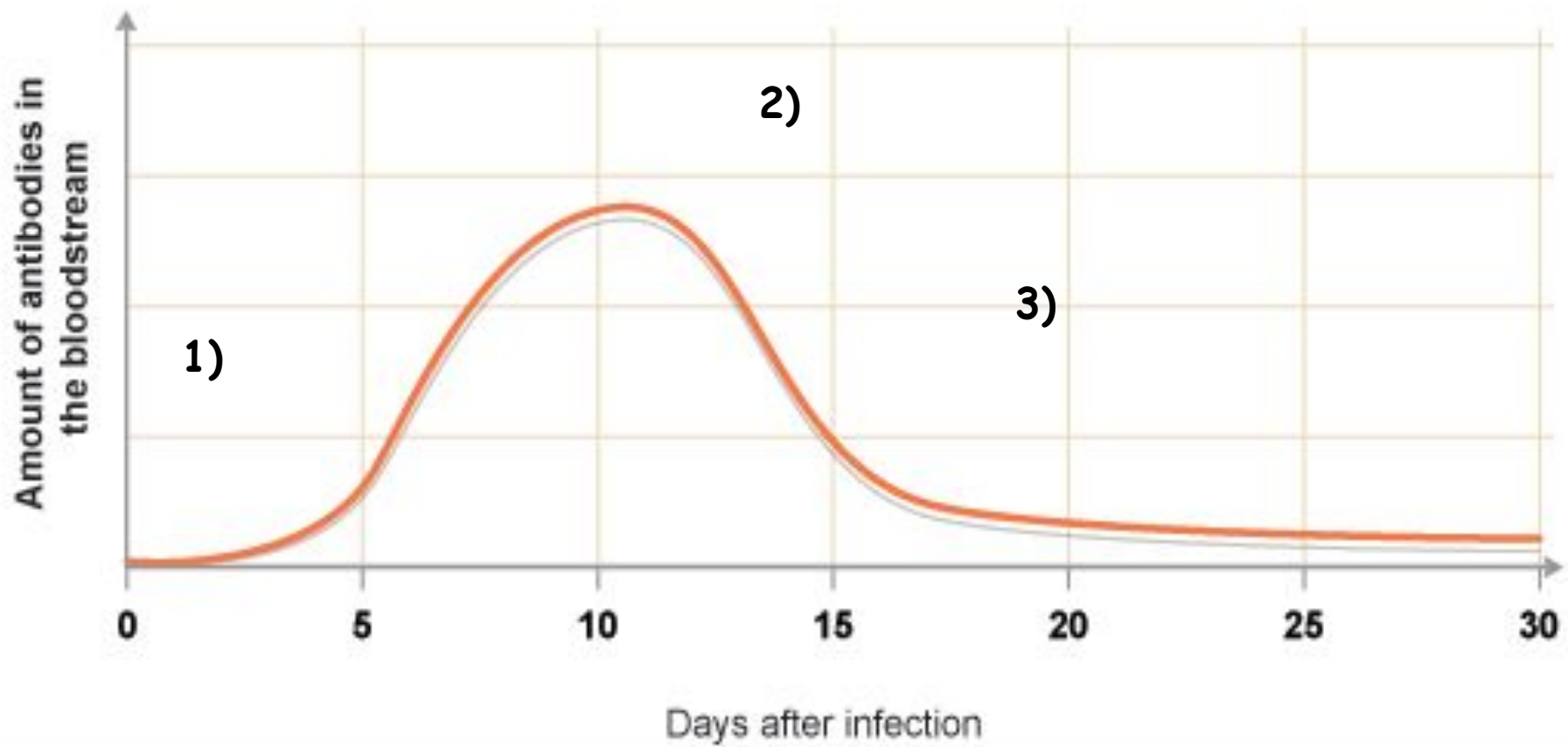


1) When you are infected by a microbe, it takes time for your body to start fighting the infection. It does this by making enough white blood cells with the correct antibody. During this time, you continue to feel unwell

2) You begin to recover when enough antibodies have been produced. After the microbes have been killed, the amount of antibodies goes down again

3) some of the white blood cells that produce the correct antibody remain in your blood.





Spread of disease

Use the pictures to create a mind map of the ways disease can spread.





Vectors (an organism that transmits disease) e.g. mosquito



Contaminated food



Break / cut in the skin



Dirty/
Contaminated
water

Spread of
disease

Bodily fluids
(droplets)



Direct
contact
(touch)



Complete the worksheet

Edward Jenner and smallpox

1. virus
2. by infecting himself with cowpox, which is similar but harmless
3. They were worried that they would develop cowpox.
4. Cowpox must have stimulated the body to make antibodies that could also destroy smallpox.

Vaccines

1. approximately 42
2. vaccine response – fewer antibodies are made, response is slower
3. Ideas include:
 - lymphocytes detect antigens on the dead or inactive pathogen; they produce a specific antibody for these antigens
 - antibodies lock onto the antigens
 - lymphocytes ‘remember’ the shape of the antigens
 - when there is a real infection due to a live pathogen entering the body, lymphocytes instantly recognise the pathogen because it has the same antigens as the vaccine
 - lymphocytes quickly make many specific antibodies
 - antibodies lock onto the pathogens and kill them before they have a chance to make you feel ill.

- **Antigen** - a toxin or other foreign substance which induces an immune response in the body, especially the production of antibodies.
- **Antibody** - protein normally present in the body or produced in response to an antigen, which it neutralises, thus producing an immune response
- **Pathogen** - a bacterium, virus, or other microorganism that can cause disease.
- **Phagocytosis** - Phagocytosis is the process by which a cell uses its plasma membrane to engulf a large particle

- **Antigen** - a toxin or other foreign substance which induces an immune response in the body, especially the production of antibodies.
- **Antibody** - protein normally present in the body or produced in response to an antigen, which it neutralises, thus producing an immune response
- **Pathogen** - a bacterium, virus, or other microorganism that can cause disease.
- **Phagocytosis** - Phagocytosis is the process by which a cell uses its plasma membrane to engulf a large particle